

Survey of Film Bulk Acoustic Resonators

MICHAEL HARRINGTON
Department of Electrical and Computer Engineering
Georgia Institute of Technology
Atlanta, United States
mharrington35@gatech.edu

Abstract-This paper will be a survey of Film Bulk Acoustic Resonators (FBARs). As wireless communication systems become more prevalent, the need for high-performance MEMS resonator devices has greatly increased. FBARs offer key metrics that make them a very appealing device for this, with a high-quality factor (Q), small form factor, and compatibility with CMOS interface circuits. These make them a popular choice for front-end applications in high-frequency filtering. This paper reviews the operating principles of FBARs, piezoelectric thin-film fabrication techniques, and approaches for achieving CMOS integration. Circuit-level interface architectures are also discussed with emphasis on potential challenges in design and fabrication.

I. INTRODUCTION

Traditionally, resonators have been designed as RCL or LC electrical circuits. These resonators face problems that in the modern world are unacceptable – poor handling at high frequencies, large size, poor nonlinearity limits, poor frequency-selectivity, etc. The demand for compact, low-loss, high-selectivity has greatly increased with a crowded electromagnetic spectrum and growing complex communication networks. Instead, acoustic resonators can meet these performance standards using mechanical vibrations that can detect microns of movement,

while being extremely small-form and having a high Q factor. Piezoelectric technologies are dominant in resonator and high-frequency MEMS applications. [1]

This paper will provide a survey on the many uses and configurations of FBARs and will be broken down into the following sections: Section II speaks on the principle of operation, Section III outlines fabrication and integration approaches, Section IV discusses circuits and system architecture for interfacing FBARs, and Section V concludes the paper.

II. PRINCIPLES OF OPERATION

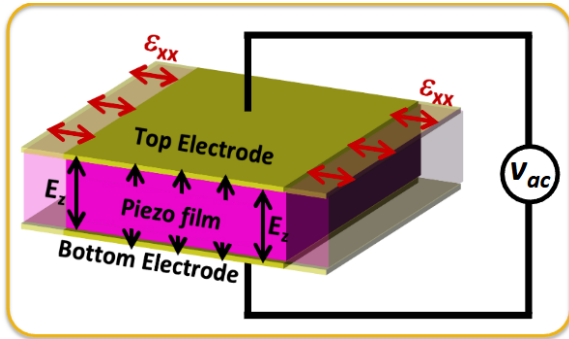
FBARs are transducers in that they convert energy into signals. They operate by converting electrical signals into mechanical acoustic waves within a thin piezoelectric film. The driving effect is known as the inverse piezoelectric effect, in which a material will deform in response to an applied electric field. The piezoelectric layer expands and contracts, launching a bulk acoustic wave normal to the film plane.

A few materials are exceptional in promoting this effect in a precise and linear manner. Specifically, natural crystals such as Quartz (SiO_2), Lithium Niobate (LiNbO_3), Gallium Arsenide (GaAs), Aluminium Nitride (AlN), Gallium Nitride (GaN), and Zinc Oxide

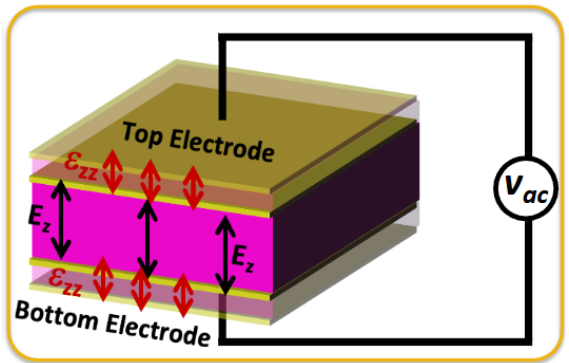
(ZnO) can also be used to facilitate piezoelectric effects in the film [1][2]. The two primary materials though are ZnO and AlN, because of their unique place in longitudinal and transverse piezoelectric coefficients and extensive research on these materials, as seen in figure 1 [1].

d_{31} and d_{33} of selected piezoelectric materials		
Material	d_{31} (pC/N)	d_{33} (pC/N)
AlN	-1.73	5.49
ZnO	-5.43	11.67
LiNbO ₃	-1	6
PZT-5H	-274	593
BaTiO ₃	-34.5	85.6

These coefficients relate to the way a mechanical deformation will occur, with d_{31} being a shear or transverse excitation, as seen in figure 2a, and d_{33} being a longitudinal excitation as seen in figure 2b [1].



a.



b.

Despite ZnO having a larger longitudinal piezoelectric coefficient and larger difference – and so should be superior in

exemplifying the mechanical deformation – AlN is typically used as the film material in FBARs. This is due to the wider band gap in the compound, superior CMOS compatibility, and less parasitic capacitance associated with the lower dielectric constant of AlN.

A typical FBAR consists of a thin piezoelectric film sandwiched between two metal electrodes and depending on the configuration some sort of confinement medium. Applying a voltage, and thereby electric field, to the electrodes produces stress in the piezoelectric layer equivalent to the piezoelectric coefficients d_{33} and d_{31} . This launches a wave whose frequency is approximately: [3][4]

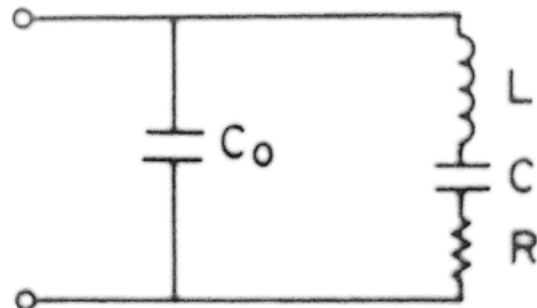
$$f_r \approx \frac{v}{2t_p}$$

where

v = acoustic velocity in piezoelectric material

t_p = piezoelectric film thickness

A piezoelectric resonator can be modeled with an LCR equivalent circuit, known as the Butterworth-Van Dyke (BVD) model. This model consists of LCR, which are motional inductance, motional capacitance, and motional resistance, and a C_0 representing the static capacitance of the electrodes as seen in figure 3 [1][5].



This model captures series and parallel resonance, which defines the stopband and passband and can be used in finding the

effective coupling factor to establish overall performance of the piezoelectric resonator.

III. Fabrication and CMOS Integration

FBARs are implemented primarily in two ways. Either as solidly mounted resonators to the substrate through acoustic reflectors (SMRs) or resonators that are free-standing. Each of these methods have their benefits and drawbacks, which are listed below:

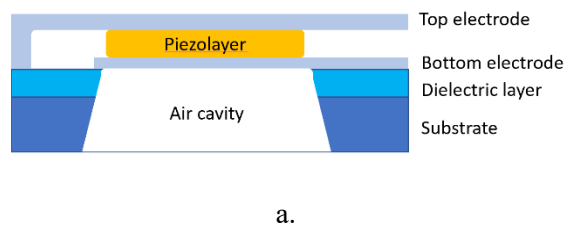
Free-Standing FBAR

- Higher Q
- Better acoustic isolation
- More difficulty fabricating

Solidly Mounted FBAR

- Very easy to implement into CMOS
- Requires acoustic mirror

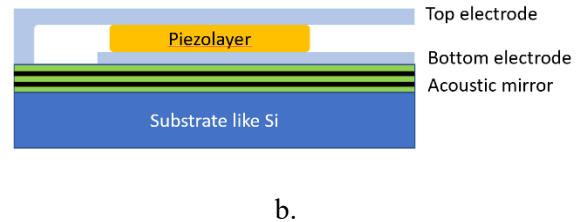
Free-standing bulk acoustic wave resonators typically implement bulk micromachining techniques, as seen in figure 4a [2].



These fabrication techniques etch out an air cavity in the Si and SiO₂ layer. This cavity is important in that when the film produces standing waves from a voltage excitement, the waves must be confined to achieve the highest possible quality factor.

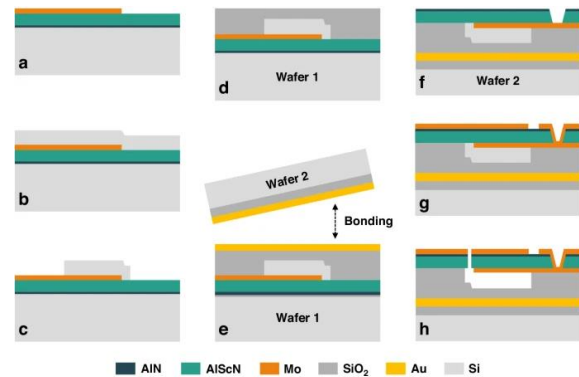
Solidly mounted bulk acoustic wave resonators instead implement a device called an acoustic mirror, typically a Bragg reflector, built on a stack. This stack mimics the acoustic

isolation that free-standing FBARs have using reflectors that reflect energy back into the resonator and prevent energy leakage, as seen in figure 4b [2].



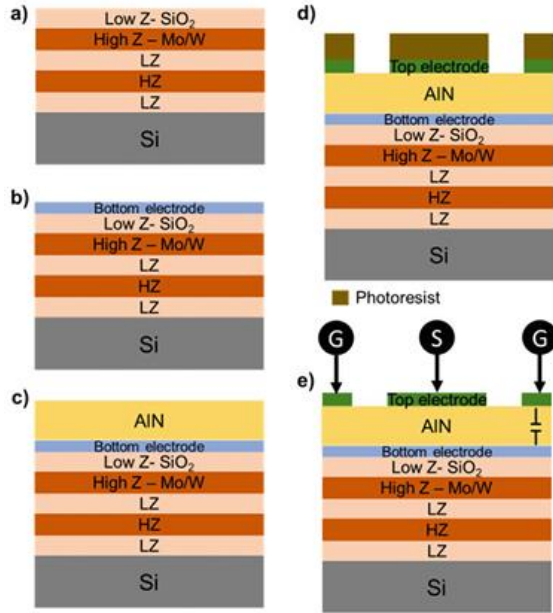
As mentioned before, the dominant film is Aluminium Nitride (AlN), and is used in manufacturing due to its low dielectric loss, wide band gap, CMOS compatibility, and strong longitudinal piezoelectric constant. The typical deposition technique for this material is reactive sputtering in Nitrogen plasma.

Though there are many techniques to creating a free-standing FBAR, a unique and quite genius process flow is seen in figure 5 below [6].



This process flow first involved the deposition of the AlN layer, then etching out the bottom electrode using Inductively Coupled Plasma on molybdenum (Mo). A cavity is created out of sacrificial Silicon before bonding an additional wafer to it via an Au-Au interface. Finally, removing the Si layer on wafer 1 and inverting the structure, and with some additional depositions, the free standing FBAR is created. [6]

The fabrication process for creating a solidly mounted resonator is more simple and uses the multilayer acoustic mirror instead of etching out a cavity, as seen in figure 6 [7].



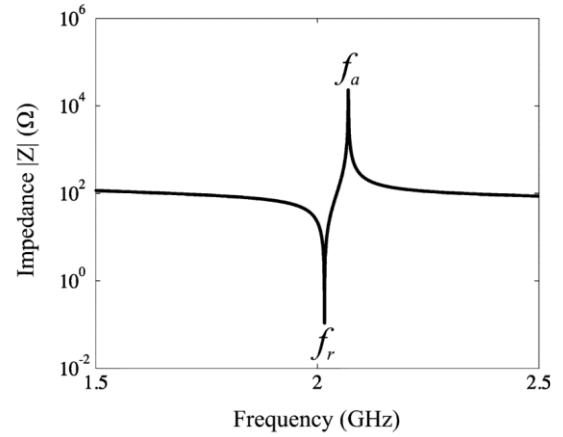
This technique has a base Silicon layer followed by an odd number of alternating deposition of low and high acoustic impedance materials. These form the acoustic reflector. Then, the bottom electrode, film, and top electrode are deposited, finalizing the SMR structure [7].

IV. CIRCUITS & SYSTEM ARCHITECTURE

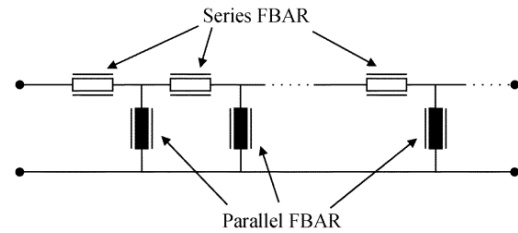
FBARs are most typically used in front-end RF systems. Two configurations that are widely used are ladder filters and lattice filters.

Before speaking on those two types of filters, it must be mentioned why the FBAR is great at creating filters in the first place, and why exactly they are so important graphically. The following figure 7 shows the electrical impedance as a function of frequency for an FBAR [8]. This plainly shows the driving point of FBARs, in that they provide an f_a that approaches infinity, and an f_r that approaches zero, and is slightly lower than f_a . This behavior

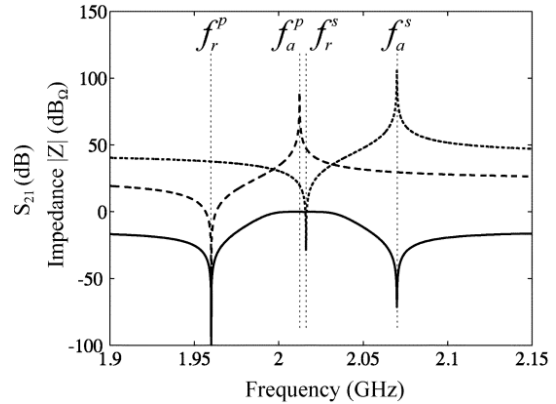
can be used in creating high Q bandpass filter, that allows for a an ideal transmission at a very, very specific frequency followed by a quick roll off at a very small Δx away from that frequency. This fact ties into why the FBAR is seen in so many front-end modules in RF applications. [8]



The ladder filter as seen in figure 7a, is made up of FBARs in series and parallel, which have each of their f_a 's and f_r 's set to be at specific locations to counteract one another, which can be seen in figure 7b [8].



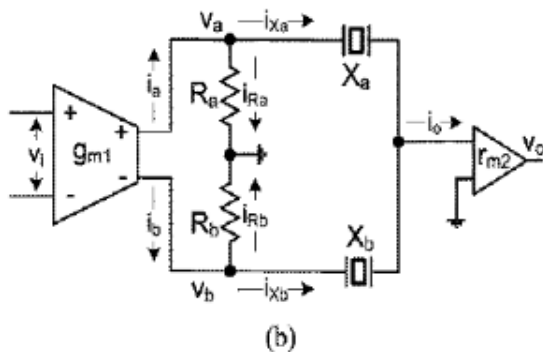
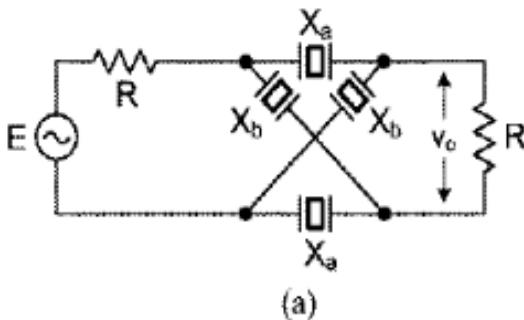
a.



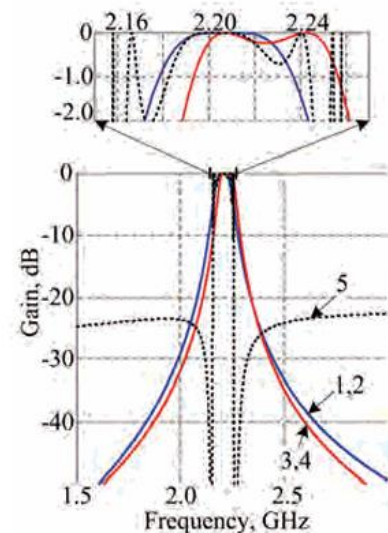
b.

This impedance to frequency figure deserves thorough explanation to fully understand the impact this has. Here, the solid line is the true line, which represents the transmission response of the filter. The dashed lines are the impedance to frequency of the FBARs that are in series or parallel, hence the belonging to f_r^p , f_a^p – these are the f_a and f_r of the parallel FBARs and f_r^s , f_a^s , the f_a and f_r of the series FBAR. The frequencies for the parallel and series FBARs are precisely set with a desired frequency in mind, with the f_a^p frequency being the beginning of the precise pass band and the f_r^s being the end of the precise pass band. This creates a center passband that is very narrow but where these reverse impedances cancel out, and large roll offs at the edges to create a more general pass band into notches on either side [1] [8].

The next circuit configuration of interest is the lattice filter, as seen in figure 8a, an additional configuration that uses less resonators but accomplishes a similar output is seen in figure 8b [9].



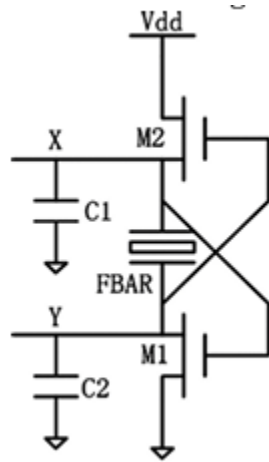
The lattice filter acts similarly to the ladder filter in that the series resonant frequency should be equal to that of the parallel resonant frequency of the other FBAR [9]. From these, the frequencies cancel out at a passband and create deep notches directly outside of the passband, as seen in figure 9 below [9].



The difference between the lattice filter and the ladder filter is that the ladder filter has a much narrower bandpass and steeper slopes outside of that band. This is due to the multiple transmission zero's in the region of the passband of the ladder filter but not of the lattice filter. But, the lattice filter will most likely contain less resonators – especially with the modified version – than that of the ladder filter, which typically has 4-8 pairs of FBARs.

As said, FBARs are typically seen in RF front end systems like duplexers or multiplexers, but they can also be used in impedance-based readout devices like sensors, or in oscillator architectures. For oscillator architectures, incorporating an FBAR into a CMOS oscillator provides sought-after characteristics, like very low power consumption for a low supply voltage and very low phase noise. The figure 10 below shows an example of implementing an FBAR

into a CMOS oscillator, in replacement of an LC tank.



This configuration allows for a very high Q with a stable resonance and lower power consumption than using an LC tank that will have a lower Q and require additional negative conductance to maintain oscillatory behavior. FBARs are not just for filters or just for RF applications, but can be used as a resonator as seen in this example or even as a sensor for mass changes, temperature changes, etc.

V. CONCLUSION

FBARs provide small-form, high-Q GHz filtering essential for modern needs. The piezoelectric effects enable precise mechanical vibrations, and fabrication techniques such as AlN sputtering and acoustic mirrors allow excellent CMOS integration. As the electromagnetic spectrum becomes increasingly crowded, these circuit architectures will become more widely use and evolve towards higher frequencies and even narrower bandwidths. Concerns such as linearity, parasitics, and integration into MEMS/CMOS modules will become even more critical as the future develops, but FBARs will play a crucial role in it.

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